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Whole Exome Sequencing Analysis

Patient ID: XX	**PIN:** XX
Age: 14 Years	**Gender:** Female
Hospital/Clinic: XX	**Sample Number:** XX
Referring Clinician: XX	**Sample Collection Date:** NA
Specimen: DNA	**Sample Received Date:** NA
	Report Date: NA

Clinical History

Proband, XX is born of a non-consanguineous marriage. XX is diagnosed with young-onset diabetes mellitus (age at onset: 12 years). On examination, BMI is 22.8 kg/m², C-peptide levels (fasting) is 0.35 ng/mL, GAD and IA-2 are negative. There is a family history of diabetes in grandparents. XX is currently being treated with insulin. Proband, XX has been evaluated for pathogenic variations.

Results

Variant of uncertain significance related to the given phenotype was identified

List of uncertain significant variant identified:

Gene	Variant*	Allele Status	Region	Disease or Disorder	Inheritance	Classification*
ABCC8 (-)	c.4195G>C **(p.Glu1399Gln)**	Heterozygous	Exon 34	Maturity-onset diabetes of the young, type 12 **(OMIM#621196)**	Autosomal Dominant	**Uncertain Significance (PM2_Supp,PP3)**

Variant Interpretation

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
ABCC8 (-)	c.4195G>C **(p.Glu1399Gln)**	Autosomal Dominant	Maturity-onset diabetes of the young, type 12 **(OMIM#621196)**	PM2_Supp,PP3	**Uncertain Significance**
Region	**Allele Status**	**Sequence Ontology**	**Variant Frequency**	**Allelic Read Depths**	
Exon 34	Heterozygous	Missense	Not observed in annotated gnomAD v4 All	Ref(C): 75, Alt(G): 57, VAF: 43.18%	
***In silico* Prediction Tools**				**Conservation Status**	
SIFT:**Da** LRT:**D**	PolyPhen2:**Da**	CADD:**U**	MT2:**D**	PhyloP: **C**	GERP++: **C**
Genomic Position			**Clinical & Literature Evidences**		
Chr11:NC_000011.10: g.17395855C>G (NM_000352.6)			-		

Abbreviation: D: Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain; **Supp:** Supporting

OMIM Phenotype: Maturity-onset diabetes of the young type 12 (OMIM#621196) is caused by heterozygous mutation in the *ABCC8* gene (OMIM*600509. Maturity-onset diabetes of the young-12 (MODY12) is a type of monogenic diabetes with the diagnostic criteria of autosomal dominant inheritance, insulin independence within 2 years of onset, at least 1 family member diagnosed with diabetes before age 25 years, and islet beta-cell dysfunction. This disease follows autosomal dominant pattern of inheritance [2].

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].
No other variant that warrants to be reported for the given clinical indication was identified.

CNV Findings

No significant Copy Number Variations (CNV) related to phenotype was detected.

Recommendations

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the blood was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS- SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not report ed. Gene list used in the assay Click here.

Sequence data attributes

Total Read Generated	**12.83 Gb**
Data ≥ Q30	**93.70%**

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non - disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions. The variants of uncertain significance and variations with high minor allele frequencies which are likely to be benign will be given upon request.

- In addition, this test does not cover detection of such structural variants in the genes *SMN1, SMN2, HBA1, HBA2, PMS2, GBA,* and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

References

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in medicine: official journal of the American College of Medical Genetics vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

This report has been reviewed and approved by:

Sivasankar.S, Ph.D Molecular Biologist	Muthukumaran. S, Ph.D Clinical Bioinformatician	Dr. Sarath R.S DM (Medical Genetics) MD DNB(Pediatrics) Consultant, Clinical Genetics	Sachin. D.Honguntikar, Ph.D, Molecular Geneticist	Dr. G. Suriyakumar Director

Appendix 1: Gene Coverage

Indication Based Analysis:

Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered
ABCB4	100.0	*ABCC8*	100.0	*ACBD6*	100.0	*ADA2*	100.0
ADAR	100.0	*ADIPOQ*	100.0	*ADRA2A*	100.0	*AEBP1*	100.0
AGPAT2	100.0	*AIP*	100.0	*AIRE*	100.0	*AKT2*	100.0
ALMS1	100.0	*AMACR*	100.0	*APOA5*	100.0	*APOE*	100.0
APPL1	100.0	*AR*	100.0	*ARHGEF15*	100.0	*ARL6*	100.0
ATM	100.0	*ATRX*	100.0	*BAZ1B*	100.0	*BBIP1*	100.0
BBS1	100.0	*BBS10*	100.0	*BBS12*	100.0	*BBS2*	100.0
BBS4	100.0	*BBS5*	100.0	*BBS7*	100.0	*BBS9*	100.0
BLK	100.0	*BLM*	100.0	*BMP2*	100.0	*BMP6*	100.0
BRCA1	100.0	*BRCA2*	100.0	*BSCL2*	100.0	*BUD23*	100.0
CARS1	100.0	*CASR*	100.0	*CAT*	100.0	*CAV1*	100.0
CBLB	100.0	*CCDC28B*	100.0	*CD274*	100.0	*CDH23*	100.0
CDKN2A	100.0	*CEL*	98.59	*CELA2A*	100.0	*CEP19*	100.0
CEP290	100.0	*CFAP418*	100.0	*CFTR*	99.98	*CIDEC*	100.0
CISD2	100.0	*CLCNKB*	100.0	*CLIP2*	100.0	*CNBP*	100.0
CNOT1	100.0	*CORIN*	100.0	*COX18*	100.0	*CP*	100.0
CPA1	100.0	*CPE*	100.0	*CTC1*	100.0	*CTLA4*	100.0
CTNNB1	100.0	*CTNS*	100.0	*CTRC*	100.0	*CYP19A1*	100.0
DCAF17	100.0	*DKC1*	100.0	*DLK1*	100.0	*DMXL2*	100.0
DNAJC21	94.99	*DNAJC3*	98.41	*DNAJC30*	100.0	*DNASE2*	100.0
DNM1L	100.0	*DNMT1*	100.0	*DUT*	100.0	*DYRK1B*	100.0
EDA	100.0	*EDA2R*	100.0	*EFL1*	100.0	*EIF2AK3*	100.0
EIF2S3	100.0	*EIF4H*	100.0	*ELMO2*	100.0	*ELN*	100.0
ENPP1	100.0	*FBN1*	100.0	*FDXR*	100.0	*FKBP6*	100.0
FLT1	100.0	*FOXC2*	100.0	*FOXP1*	100.0	*FOXP3*	100.0
FOXRED1	100.0	*FXN*	100.0	*GATA3*	100.0	*GATA6*	100.0
GCGR	100.0	*GCK*	100.0	*GFM2*	100.0	*GJA1*	100.0
GJB3	100.0	*GJB4*	100.0	*GLIS3*	100.0	*GLRX5*	100.0
GOSR2	100.0	*GPD2*	100.0	*GPR101*	100.0	*GPR35*	100.0
HAMP	100.0	*GTF2IRD1*	100.0	*HERC2*	100.0	*GYG1*	100.0
HJV	100.0	*HBB*	100.0	*HLA-DQB1*	100.0	*HFE*	100.0
HMGA1	100.0	*HMGA2*	100.0	*HNF1A*	100.0	*HLA-DRB1*	100.0
HNF4A	100.0	*HR*	100.0	*HSPB8*	100.0	*HNF1B*	100.0
IER3IP1	100.0	*IFIH1*	100.0	*IFT172*	100.0	*IARS1*	100.0
IFT74	100.0	*IGF1R*	100.0	*IGF2BP2*	100.0	*IFT27*	100.0
IL18BP	100.0	*IL2RA*	100.0	*IL6*	100.0	*IGHG2*	100.0
INSR	100.0	*IRS1*	100.0	*IRS2*	100.0	*INS*	100.0
KCNJ11	100.0	*KCTD1*	100.0	*KDSR*	100.0	*ITCH*	100.0
KRAS	100.0	*LAMB1*	100.0	*LEMD3*	100.0	*KLF11*	99.94
LEPR	100.0	*LHX1*	100.0	*LIG4*	100.0	*LEP*	100.0
LIPC	100.0	*LIPE*	100.0	*LMF1*	100.0	*LIMK1*	100.0
LMNB2	100.0	*LRBA*	100.0	*LRP6*	100.0	*LMNA*	100.0
LZTFL1	100.0	*MAFA*	100.0	*MAGEL2*	100.0	*LSM11*	100.0
MAPK8IP1	100.0	*MC4R*	100.0	*MEF2A*	100.0	*MANF*	100.0
MEN1	100.0	*METTL27*	100.0	*MIA3*	100.0	*MEG3*	100.0
MKKS	100.0	*MKRN3*	100.0	*MKS1*	100.0	*MICU1*	100.0
MMP14	100.0	*MMP2*	100.0	*MOG*	100.0	*MLXIPL*	99.58
MST1	100.0	*MTNR1B*	100.0	*NAF1*	100.0	*MPV17*	100.0
NDUFA11	100.0	*NDN*	100.0	*NDP*	100.0	*NARS2*	100.0
NDUFAF3	100.0	*NDUFA6*	100.0	*NDUFAF1*	100.0	*NDUFA1*	100.0
NDUFB10	100.0	*NDUFAF4*	100.0	*NDUFAF5*	100.0	*NDUFAF2*	100.0
NDUFB9	100.0	*NDUFB11*	100.0	*NDUFB2*	100.0	*NDUFAF8*	100.0
NDUFS4	100.0	*NDUFS1*	100.0	*NDUFS2*	100.0	*NDUFB3*	100.0
NDUFV1	100.0	*NDUFS6*	100.0	*NDUFS7*	100.0	*NDUFS3*	100.0
NEUROG3	100.0	*NDUFV2*	100.0	*NEK8*	100.0	*NDUFS8*	100.0

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NPAP1	100.0	*NHP2*	100.0	*NOP10*	100.0	*NEUROD1*	100.0
NR3C1	100.0	*NPHP1*	100.0	*NPHP3*	100.0	*NOTCH3*	100.0
OPA1	100.0	*NSMCE2*	100.0	*NUBPL*	100.0	*NPM1*	100.0
PAX4	100.0	*PALB2*	100.0	*PALLD*	100.0	*OCA2*	100.0
PDCD1	100.0	*PCBD1*	100.0	*PCNT*	100.0	*PARN*	100.0
PEX10	100.0	*PDE4D*	100.0	*PDX1*	100.0	*PCYT1A*	100.0
PLAAT3	100.0	*PEX6*	100.0	*PI4KA*	100.0	*PEX1*	100.0
PNPLA2	100.0	*PLAGL1*	100.0	*PLCD1*	100.0	*PIK3R1*	100.0
POLD1	100.0	*PNPLA6*	100.0	*POC1A*	100.0	*PLIN1*	100.0
PPARG	100.0	*POLG*	100.0	*POLG2*	100.0	*POLA1*	100.0
PRKACA	100.0	*PPP1R15B*	100.0	*PPP1R3A*	100.0	*POLR3A*	100.0
PRSS2	100.0	*PRKAR1A*	100.0	*PROK2*	100.0	*PPP2R5C*	100.0
PTRH2	100.0	*PSTPIP1*	100.0	*PTF1A*	100.0	*PRSS1*	100.0
RETN	100.0	*PWRN1*	100.0	*RABL3*	100.0	*PTPN1*	100.0
RTL1	100.0	*RFC2*	100.0	*RFX6*	100.0	*RAC1*	100.0
SCAPER	100.0	*RNASEH2C*	100.0	*RRM2B*	100.0	*RNASEH2A*	100.0
SIM1	100.0	*SAMHD1*	100.0	*SARS2*	100.0	*RTEL1*	100.0
SLC25A4	100.0	*SCLT1*	100.0	*SDCCAG8*	100.0	*SBDS*	100.0
SLC37A4	100.0	*SIN3A*	100.0	*SLC12A3*	100.0	*SEMA4D*	100.0
SNRPN	100.0	*SLC29A3*	100.0	*SLC2A2*	100.0	*SLC19A2*	99.8
STAT1	100.0	*SLC40A1*	100.0	*SMAD4*	100.0	*SLC30A8*	100.0
STX1A	100.0	*SPI1*	100.0	*SPINK1*	100.0	*SMPD4*	100.0
TERC	100.0	*STAT3*	100.0	*STOX1*	99.33	*SPOUT1*	100.0
TINF2	100.0	*TBL2*	100.0	*TCF4*	100.0	*STUB1*	100.0
TMEM270	100.0	*TERT*	100.0	*THRB*	100.0	*TCF7L2*	100.0
TTC7A	100.0	*TKT*	100.0	*TLR8*	100.0	*TIMMDC1*	100.0
TYMS	100.0	*TP53*	100.0	*TRMT10A*	100.0	*TMEM126B*	100.0
USP8	100.0	*TTC8*	100.0	*TTPA*	100.0	*TRPV6*	100.0
WRAP53	100.0	*UBR1*	100.0	*USB1*	100.0	*TWNK*	100.0
ZBTB20	100.0	*VPS37D*	100.0	*WDPCP*	100.0	*USP48*	100.0
ZNF668	100.0	*WRN*	100.0	*XRCC4*	100.0	*WFS1*	100.0
ZFP57	100.0	*ZFYVE26*	100.0	*YIPF5*	100.0		